



ASSAM SCIENCE AND TECHNOLOGY UNIVERSITY GUWAHATI

**Course Structure and Syllabus
(From Academic Session 2018-19 onwards)**

**B. TECH
POWER ELECTRONICS AND INSTRUMENTATION ENGINEERING
8th SEMESTER**



ASSAM SCIENCE AND TECHNOLOGY UNIVERSITY
Guwahati
Course Structure

(From Academic Session 2018-19 onwards)
B.Tech 8th Semester: Power Electronics and Instrumentation Engineering

Semester VIII/ B.TECH/PEI

Sl. No	Sub-Code	Subject	Hours per Week			Credit	Marks	
			L	T	P		C	CE
Theory								
1	PI181801	Power Electronics Circuit and Application	3	0	0	3	30	70
2	PI181802	Analytical Instrumentation	3	0	0	3	30	70
3	PI1818PE3*	Program Elective-3	3	0	0	3	30	70
4	PI1818OE3*	Open Elective-3	3	0	0	3	30	70
Practical								
1	PI181813	Virtual Instrumentation lab	0	0	2	1	15	35
2	PI181821	Project-2	0	0	12	6	100	50
3	PI181822	General Viva	0	0	0	2	-	100
TOTAL			12	0	14	21	235	465
Total Contact Hours per week: 26								
Total Credit: 21								

PROGRAM ELECTIVE – 3		
Sl.No.	Subject Code	Subject
1	PI1818PE31	Advanced Process Control
2	ECE1818PE33	Biomedical Electronics
3	PI1818PE3*	Any other subject offered from time to time with the approval of the University

OPEN ELECTIVE – 3		
Sl.No.	Subject Code	Subject
1	PI1818OE31	Biomedical Signal Processing
2	PI1818OE3*	Any other subject offered from time to time with the approval of the University

Detail Syllabus:

Course Code	Course Title	Hours per week L-T-P	Credit C
PI181801	Power Electronics Circuit and Application	3-0-0	3

Course Outcomes (CO):

At the end of the course, the students will be able to:

CO1: To get an overview of electric drive system.

CO2: Control of dc and ac drive system with Power Electronics based circuits.

CO3: Control of electric devices in process plant.

CO4: The fundamentals of dc power supply including uninterrupted power supply system.

CO5: Understanding the concept of power factor improvement.

MODULE 1: Introduction

Concept of drives; factors influencing the selection of motors & their control; classification of loads; dynamics of load-motor combination; four –quadrant operation of motor loads.

MODULE 2: DC Drives

Concept of Electric Drive, DC Drives, Single phase DC Drives, Three phase DC Drives, Chopper Drives.

MODULE 3: AC Drives

Induction- Motor Drives, Speed Control of Three- Phase Induction Motors, Synchronous Motor Drives.

MODULE 4: Industrial Application

Different types of steel mills and drives used in steel mills; types of paper mill drive; comparison between line shaft and sectional drives; cement factory, textile mills, sugar mills and coal mine drives, machine tool applications.

MODULE 5: Electric Traction

Introduction to electric traction system, speed time curves & speed distance curves, choice of traction system in India.

MODULE 6: Electric Heating & Welding

Electric heating-relative cost, advantages, types & applications; transfer of heat; heating and cooling curves.

MODULE 7: Power Supplies

D.C power supply; SMPS DC power supply; resonant DC power supply; bidirectional power supplies; AC power supplies; UPS configuration; SMPS ac power supplies.

MODULE 8: Power Factor Conditioning

Effect of Poor Power factor, Methods of Reactive Power Compensation, Static VAR Compensator.

REFERENCES:

1. Utilization of Electric Power-Suryanarayan N.V
2. A First Course on Electric Drives- S.K. Pillai

3. Utilization of Electric Energy- E.O.Taylor
4. Electric Traction- A.T. Dover
5. Generation, Distribution and Utilization of Electric Energy- C.L. Wadhwa.
6. Power Electronics-Dr. P.S. Bimbhra

Course Code	Course Title	Hours per week L-T-P	Credit C
PI181802	Analytical Instrumentation	3-0-0	3

Course Outcomes (CO):

At the end of the course, the students will be able to

CO1: Select the required instruments for a particular analysis with idea of its merits, demerits and limitations.

CO2: Acquire knowledge about the interaction of electromagnetic radiations with matter and apply analytical techniques to accurately determine the elements present in the given sample.

CO3: Learn specific technique employed for monitoring different pollutants in air and water.

CO4: To apply the knowledge of chromatography to separates the constituents from a complex mixture in real time industrial environments.

CO5: Understand the working principles of different gas analyzers, electrodes, NMR Spectrometers and radiations techniques.

CO6: Acquire knowledge about humidity, density and moisture measurements.

MODULE 1: Measurement of Density, Humidity and Moisture

Measurement of density- Float type, Different types of Hydrometers, Ultrasonic method, and Radiation method. Measurement of Humidity-Dry and wet bulb psychrometer, capacitive method, Different types of Hygrometers. Measurement of moisture in gases, liquids and solids.

MODULE 2: Chromatography

Different techniques – Gas chromatography, Liquid chromatography, principles, types and applications, Detectors.

MODULE 3: Industrial Gas Analyzers and Pollution Monitoring Instruments

Types of gas analyzers – Oxygen, NO₂ and H₂S types, IR analysers, thermal conductivity analyser. Air pollution due to carbon monoxide, hydrocarbons, nitrogen oxides, sulphur dioxide and their estimation techniques, Dust and smoke measurements.

MODULE 4: pH METERS AND DISSOLVED COMPONENT ANALYSERS

Principle of pH measurement, glass electrodes, hydrogen electrodes, reference electrodes, selective ion electrodes, ammonia electrodes, biosensors, dissolved oxygen analyser, Sodium analyser, Silicon analyser.

MODULE 5: Nuclear Magnetic Resonance and Radiation Techniques

NMR – Basic principles – NMR spectrometer applications. Introduction to Mass spectrophotometers, nuclear radiation detectors-GM counter, Proportional counter. Introduction to X-ray spectroscopy.

MODULE 6: Spectrophotometry

Special methods of analysis – Beer-Lambert law – Colorimeters – UV-Visible spectrophotometers – Single and double beam instruments –Sources and detectors-- IR Spectrophotometers – Types – Sources and detectors – FTIR spectrophotometers – Atomic absorption spectrophotometer-Flame emission spectrophotometers – sources of flame photometry-applications

REFERENCES:

1. G.W. Ewing, 'Instrumental Methods of Analysis', McGraw Hill, 1992.
2. R.K.Jain, Mechanical and Industrial Measurements, Khanna Publishers, New Delhi, 1999
3. H.H. Willard, L.L. Merritt, J.A. Dean, F.A. Settle, 'Instrumental Methods of Analysis', CBS publishing & distribution, 1995.
4. Robert D. Braun, 'Introduction to Instrumental Analysis', McGraw Hill, Singapore, 1987.
5. R.S. Khandpur, 'Handbook of Analytical Instruments', Tata McGraw Hill publishing Co. Ltd., 2003.
6. Liptak, B.G, Process Measurement and Analysis, Chilton Book Company, 1995
7. K. Krishnaswamy- Industrial Instrumentation (New Age)

Course Code	Course Title	Hours per week L-T-P	Credit C
PI1818PE31	Advanced Process Control	3-0-0	3

Course Outcomes (CO):

At the end of the course the students will be able to:

CO1: Understand the basic concepts of advanced process control schemes

CO2: Explain the working of distributed control system

CO3: Elaborate the use of artificial intelligence techniques in process control.

CO5: Explain the concept of digital control system.

CO4: Explain the fundamental concepts of PLC.

MODULE 1: Introduction to Advanced Control Schemes

Cascade, Feed-forward, Feed-forward plus Feedback, Ratio control, Inferential control, Dead time and Inverse response compensation, Adaptive control, Model reference adaptive control, Self-tuning regulator Interactions and Decoupling of Control Loops: Design of cross controllers and selection of loops using Relative Gain Array

MODULE 2: Distributed Control System (DCS)

Evolution and advantages of computer control, Configuration of Supervisory, Direct digital control (DDC) and DCS.

MODULE 3: Artificial Intelligence in Process Control

Expert systems, Neural networks, Fuzzy logic, Neuro Fuzzy, Genetic algorithm, Virtual instrumentation.

MODULE 4: Programmable Logic Controllers

Comparison with hard wired relay and semiconductor logic, Hardware, Ladder diagram programming, Case studies, Introduction to CPLD, SPLD, FPGA

MODULE 5: Digital Control

Sampling and reconstruction, Discrete systems analysis, Stability and controller design using z transform and difference equations, Smoothing filter realization using difference equations

TEXTBOOKS:

1. Stephanopoulos, G., Chemical Process Control, Prentice–Hall of India Private Limited (1983).
2. Liptak, B.G., Instrument Engineers Handbook, Chilton Book Company (1994).

REFERENCES:

1. Process Control -Surekha Bhanot (OXFORD)
2. Johnson, C.D., Process Control Instrumentation Technology, Prentice–Hall of India Private Limited (2007).
3. Zaidi, A., SPC Concepts, Methodologies and Tools, Prentice–Hall of India Private Limited (1995).

Course Code	Course Title	Hours per week L-T-P	Credit C
ECE1818PE33	Biomedical Electronics	3-0-0	3

Course Outcomes (CO):

At the end of this course students will be able to

CO1: Explain the principle of physiological systems and biomedical signals.

CO2: Apply electronic systems in biological and medical applications.

CO3: Develop the skills necessary to communicate findings and interpretations from medical imaging and recording systems.

CO4: Explain the use of therapeutic equipment.

MODULE 1: Physiological Systems and Signals

Biology of the heart, circulatory and respiratory systems, auditory systems, physiology of nerve and muscle cells, fundamental organization of brain and spinal cord.

MODULE 2: Biosignals

Origin of bioelectric signals, electrocardiogram (ECG), phonocardiogram (PCG), encephalogram (EEG) and electromyogram (EMG). Spectral characteristic of biosignals.

MODULE 3: Physiological Transducers

Electrodes: silver-silver chloride electrodes, electrodes for ECG, EEG, EMG, Microelectrodes. Performance characteristics of transducers, classification of transducers based on Electrical principle involved: Resistive position transducer, resistive pressure transducer, inductive pressure transducer, capacitive pressure transducer; Self generating inductive transducer: linear variable differential transformer (LVDT), Piezoelectric Transducer. Transducers for body temp measurement, photoelectric transducers, pH measurement.

MODULE 4: Recording Systems

Preamplifier, Signal conditioning: Differential amplifier, current to voltage converter, instrumentation amplifier; biomedical filters: LPF, HPF, bandpass, band stop (Notch filter); 4-20ma transmitter, source of noise in low level measurement, Recording systems for ECG, PCG, EEG and EMG.

MODULE 5: Medical Imaging Systems

X-ray imaging, Computed tomography, ultrasonic imaging systems, Magnetic resonance imaging system, thermal imaging systems.

MODULE 6: Therapeutic Equipment

Cardiac pacemaker, cardiac defibrillators, haemodialysis machine, lithotriptors, ventilators, bionic ear.

REFERENCES:

1. L. Cromwell, F. J. Weibell, E.A. Pfeiffer. "Biomedical Instrumentation and Measurement" Pearson Education, 2003
2. R.S. Khandpur, "Handbook of Biomedical Instrumentation"
3. TATA McGRAW HILL, 2005
4. J. Enderle, S. Blanchard, J. Bronzino. "Introduction to Biomedical Engineering" Academic Press, 2000

Course Code	Course Title	Hours per week L-T-P	Credit C
PI1818OE31	Biomedical Signal Processing	3-0-0	3

Course Outcomes (CO):

At the end of the course the students will be able to

CO1: The student will be able to model a biomedical system.

CO2: The student will be able to understand various methods of acquiring bio signals.

CO3: The student will be able to understand various sources of bio signal distortions and its remedial techniques.

CO4: The students will be able to analyze ECG and EEG signal with characteristic feature points.

CO5: The student will have a basic understanding of diagnosing bio-signals and classifying them.

MODULE 1: Introduction to Bio-medical Signals

Classification, Acquisition and Difficulties during Acquisition, Basic of ECG, EEG, EMG & ERG, Role of Computers in the Analysis, Processing, Monitoring & Control and image reconstruction in bio-medical field.

MODULE 2: ECG

ECG data acquisition, ECG Parameters & their estimation, Use of multiscale analysis for ECG parameters estimation-Noise & Artifacts Measurement of Amplitude and Time Intervals, QRS Detection (Different Methods), ST Segment Analysis, Removal of Baseline Wander and Power line Interferences, Arrhythmia Analysis, Portable Arrhythmia Monitors.

MODULE 3: Data Reduction

Lossless & lossy-Heart Rate Variability, Time domain measures, heart rhythm representation, spectral analysis of heart rate variability, interaction with other physiological signals, Turning point algorithm, AZTEC Algorithm, Fan Algorithm, Huffman and Modified Huffman Coding, Run Length Coding.

MODULE 4: EEG

Neurological Signal Processing, EEG characteristic, recording techniques, Sleep EEG, Dynamics of Sleep/ Wake transition, Study of pattern of brain waves, Epilepsy-Transition, Detection and Estimation, linear prediction theory, Brain Computer interface, artifacts in EEG & their characteristics and processing, model based spectral analysis, EEG segmentation, joint time frequency analysis, correlation analysis of EEG channels, coherence analysis of EEG channels.

MODULE 5: EP Estimation

By Signal Averaging, Adaptive Filtering, General Structures of Adaptive Filters, LMS Adaptive Filter, Adaptive Noise Cancelling, Wavelet Detection, Introduction, Detection by structural features, Matched Filtering, Adaptive Wavelet Detection, Detection of Overlapping Wavelets.

REFERENCES:

1. Rangaraj M. Rangayyan, "Biomedical Signal Analysis A Case Study Approach", John Wiley and Sons Inc.

2. Willis J. Tomkin, "Biomedical Digital Signal Processsing", PHI.
3. D.C. Reddy, "Biomedical Signal Processing", McGraw Hill.
4. Metin Akay, "Nonlinear Biomedical Signal Processing, Fuzzy Logic, Neural Networks and New algorithms", John Wiley and Sons.
5. A.C. Guyton, "Textbook of Medical Physiology" Prism Books (PVT) Ltd.
6. Crommwell, Weibel and Pfeifer, "Biomedical Instrumentation and Measurement", PHI.
7. John G. Webster, "Medical Instrumentation Application and Design", John Wiley & Sons Inc.
8. John L Semmlow, "Bio-signal and Biomedical Image Processing", Marcel Dekker.
9. R.S.Khandpur, "Handbook of Biomedical Instrumentation", Tata McGraw Hill, New Delhi, 2003, Edition-II

Course Code	Course Title	Hours per week L-T-P	Credit C
PI181813	Virtual Instrumentation lab	0-0-2	1

Course Outcomes (CO):

At the end of the course the students will be able to:

CO1: Understand Fundamentals of Virtual Instrumentation and control

CO2: Design virtual controllers

CO3: Apply the knowledge for real time application

LIST OF EXPERIMENTS:

SL No	Experiment	Aim
1	Creating Virtual Instrumentation for simple application	To understand the principles of Virtual Instrumentation (VI) and learn the basics for creating Virtual Instrument.
2	Graphical Programming basic tools	To understand the principles of Virtual Instrumentation Programming and learn the basics of different Loops, Charts, Clusters and Graphs
3	Data Acquisition through Virtual Instrumentation	To understand the principles of Data acquisition through Virtual Instrumentation presenting the data in real time.
4	Real Time Temperature Control	To design an On-Off controller for a real temperature process using VI
5	Design of PID Controller	To design PID controller and obtain the step input response with three different PID configurations.
6	Design of Fuzzy Logic controller	To design a fuzzy logic controller and obtain the step input response.
7	Real time DC motor speed controller using Virtual controller	To design a virtual controller for real time DC motor speed controlling.

Equipment Required:

1. PC with National Instruments LabVIEW 2008 or above version
2. LabVIEW DAQmx driver
3. DAQ system
